

# RAJVANSH MALHOTRA

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## EDUCATION

### National Institute of Technology, Kurukshetra

B.Tech, Computer Engineering

Aug 2023 – Jun 2027

## PROFESSIONAL EXPERIENCE

### Samvedya AI, in collaboration with AIIMS Delhi

Mar 2025 – May 2025

AI/ML Intern

- **Developed a LLM Pipeline** to convert PNG-based digital designs into SVG format, integrating image analysis, LLM orchestration, and API pipelines to automate a visual abstraction workflow.
- Delivered a functional MVP within 3 weeks using Python and modern ML techniques, streamlining visual abstraction for clinical research.
- Engineered a **RAG pipeline** to generate context-aware answers in a chat-based system, achieving **80% accuracy** on clinically relevant visual abstracts at AIIMS Delhi.

### National Institute of Technology, Kurukshetra

Jun 2024 – Dec 2024

Machine Learning Research Intern

- Engineered features through **extracting image features with VGG19 in TensorFlow** and **selecting top 1,000 impactful features via a Random Forest classifier**.
- **Improved model accuracy from 60% to 82% by training a Vision Transformer** on the dataset.
- Optimised predictions by integrating Golden Ratio + eye-spacing calculations via OpenCV.
- Achieved state of the art accuracy with a custom ANN on structured data, reducing training time and inference costs.

## PROJECTS

### Awaaz

May 2026

Rumik AI Hackathon - TOP 50

- Reduced perceived voice AI latency from **46s to <3s** (first audio chunk) by profiling and eliminating bottlenecks across **12 targeted optimizations**, including async parallelism, connection reuse, model swaps, and streaming architecture redesign.
- Cut nearly **500ms** from the critical inference path by decoupling **LLM inference** from **speaker identification** using **asyncio.gather**, triggering LLM execution immediately after transcript availability while speaker recognition ran concurrently.
- Eliminated repeated **TLS handshake overhead** (~400ms/request) by replacing per-request TTS client creation with a persistent module-level **httpx.AsyncClient**, significantly improving throughput and response consistency.
- Achieved real-time progressive audio playback by replacing a multi-step HTTP approval pipeline with a persistent **WebSocket streaming architecture**, enabling token streaming from LLM → live TTS chunk synthesis → immediate browser playback via the **Web Audio API**.
- Improved **Hinglish speech transcription accuracy** and removed retry overhead by replacing **Groq Whisper** with **Sarvam Saaras v3 Codemix STT**, optimized specifically for Roman-script Hindi-English code-mixed speech.
- Reduced **LLM output token usage by 60%** (512 → 200 tokens) by redesigning the model role from full conversational generation to lightweight **emotion tagging**, proportionally accelerating both inference and downstream TTS synthesis.
- Built a **personality-driven voice personalization system** that encodes user traits (energy, tone, language preference) into structured onboarding profiles, dynamically generating optimized LLM system prompts and TTS voice descriptions with near-zero per-request overhead.

### Battery Health Optimization

Mar 2026

Research Work – IEEE

- Deployed a **Physics-Informed Residual Neural Network (RPNN)** on an **ESP32 microcontroller** for real-time **SOC and SOH estimation**, embedding Coulomb counting differential equations directly into the training loss to enforce physically consistent predictions.
- Trained with a three-component loss combining **data-driven MSE**, **ODE physics residual via automatic differentiation**, and boundary constraints enforcing SOC within [0, 100]%, optimized using **AdamW** with physics loss weight  $\lambda = 0.1$ .
- Compressed the PyTorch RPNN into a **C++ TinyML engine for edge deployment**, eliminating cloud dependency for safety-critical battery diagnostics in **automotive applications**.

- Engineered **optimum\_ReAct**, a **custom agentic AI library from scratch**; deployed **3 concurrent LLM agents** (ATLAS, ORBITAL-1, ANALYST-1) with isolated context windows and role-specific system prompts, inspired by Palantir’s Gotham.
- Designed **HybridMemory** combining **BM25 retrieval**, **sentence-transformer semantic search**, and a weighted knowledge graph with multi-factor eviction (importance, recency decay, access frequency, connection weight).
- Deployed **production backend on AWS EC2** with Docker containerisation, asyncio Semaphore concurrency controls, and **exponential backoff retry (15/30/60s)** for Groq API fault tolerance.
- Shipped end-to-end **MLOps pipeline via GitHub Actions CI/CD**.

- Fine-tuned **Meta LLaMA-3 8B** using **QLoRA (Quantized LoRA)** via Unsloth for efficient low-resource conversational training.
- Applied **LoRA adapters** to key transformer components (q\_proj, v\_proj) with **r=64 and 4-bit quantisation** to reduce memory usage while preserving performance.
- Tuned generation outputs using **temperature and top\_p sampling**.

TECHNICAL SKILLS

|            |                                                                                                                                                                      |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages  | Python, C/C++                                                                                                                                                        |
| ML / AI    | Machine Learning, Deep Learning, NLP, Large Language Models, LLM Fine-tuning, RAG, Prompt Engineering, Scikit-learn, TensorFlow, PyTorch, Pandas, Matplotlib, OpenCV |
| DevOps     | AWS EC2, Docker, GitHub Actions, MLflow, Apache Airflow, ETL, SQL                                                                                                    |
| Coursework | Data Structures & Algorithms, OS, DBMS, Computer Networks, Computer Architecture, Software Engineering, Artificial Intelligence                                      |

ACHIEVEMENTS

- **Amazon ML Challenge:** Directed a high-performing team to the top **97.2 percentile**; reduced SMAPE from 100 to 48 within 2 days using targeted ML strategies.
- **Problem Solving:** Solved 300+ problems on LeetCode and other coding platforms.
- **Achieved 1 silver medal and 3 gold medals** in Taekwondo .